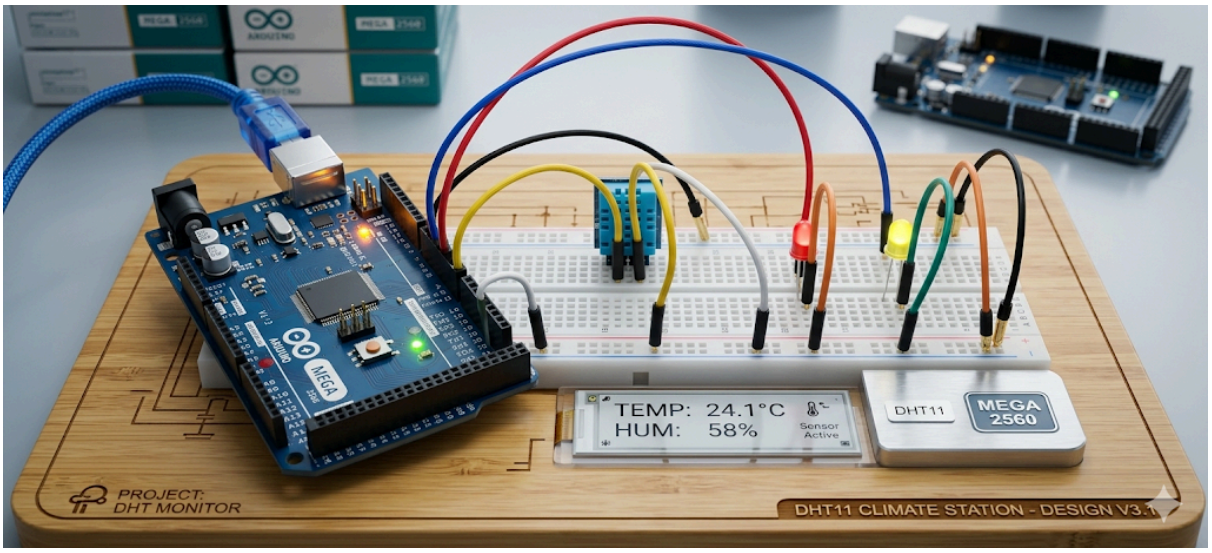


Project Title: Design and Implementation of a Dual-LED Temperature Monitoring and Alert System

University Malaysia Terengganu

Course: Elektronik and Instrumentation

Project Type: Personal Project

Author: Lim Yong Ping

Date: 14 March 2026

Abstract

This project presents the design and implementation of a temperature monitoring system using an Arduino Mega 2560 and a DHT11 sensor. The system continuously measures ambient temperature and provides visual alerts through a dual-LED architecture when predefined thresholds are exceeded. Experimental testing demonstrates that the prototype can successfully detect temperature changes and activate warning indicators at 32°C and 34°C respectively.

Introduction

Temperature monitoring is a critical component in various modern applications, ranging from industrial safety and server room maintenance to smart home automation. Prompt detection of abnormal temperature changes is essential to prevent equipment damage and ensure environmental safety.

This project focuses on the design and implementation of a real-time ambient temperature monitoring and alert system. Utilizing an Arduino Mega 2560 microcontroller and a DHT11 sensor, the system is engineered to provide immediate visual feedback through a Dual-LED architecture.

The primary objectives of this project are:

- To continuously capture and process ambient temperature data using the DHT11 sensor.
- To implement a multi-level decision logic that categorizes the severity of the environment.
- To develop a reliable visual alert mechanism that triggers a Warning state at 32°C and a Critical Alert state at 34°C.

By exploring both hardware integration and software programming, this project aims to demonstrate a practical and stable approach to automated environmental monitoring.

Components Used

1. Arduino Mega 2560 (Microcontroller)

2. DHT11 Sensor (Temperature and humidity sensor)

3. 2 × LEDs (Red and Yellow, for visual warnings)

4. Breadboard (Circuit prototyping)

5. Jumper wires (Circuit connections)

System Architecture

DHT11 Sensor

↓

Arduino Mega 2560

↓

Temperature Processing

↓

Decision Logic

↙ ↘

Warning LED Alert LED

Module Function：

Module	**Function**
DHT 11Sensor	Measure temperature
Arduino Mega 2560	Process sensor data
Decision logic	Compare current temperature with predefined thresholds
LED Indicators	Provide visual warning signals

Circuit Design

1.DHT11 Sensor Wiring:

- VCC: 5V
- GND: GND
- DATA (Signal): Digital Pin 2

2.Output Wiring:

- Warning LED (LED 1): Positive to Pin 7, Negative to GND
- Alert LED (LED 2): Positive to Pin 8, Negative to GND

Software Design

Software Algorithm:

1. Initialize the DHT11 sensor

2. Read temperature data from the sensor

3. Compare temperature with predefined thresholds

4. Activate LEDs based on temperature level

5. Repeat the process every 2 seconds

Decision logic

Temperature	**Action**
< 32°C	Normal
32–34°C	LED1 ON+LED 2 OFF
> 34°C	LED1 + LED2 ON

Alert System

Testing Results:

Temperature	LED1	LED 2	Status
26°C	OFF	OFF	Normal
33°C	ON	OFF	Warning
34°C	ON	ON	Alert

Table 1: System Response under Different Temperatures

Testing Setup: To validate the system's responsiveness, body heat (holding the sensor) was utilized to simulate a gradual temperature rise to breach the 32°C Warning threshold. Subsequently, a hairdryer was used to apply targeted heat to exceed the 34°C Critical Alert threshold.

Troubleshooting - Hardware Issue 1: Sensor Wiring Error

- **Problem:** Initially, the system exhibited erratic behavior where output components stayed continuously active, and the DHT11 sensor began overheating. The Serial Monitor returned "NaN" (Not a Number) reading errors.
- **Root Cause Analysis:** A hardware inspection revealed that the Signal, VCC, and GND pins on the DHT11 sensor were incorrectly wired, causing an internal short circuit that halted the execution of the main code logic.
- **Solution & Methodology:** An "Isolation Test" was immediately conducted. A pure 'Blink' code was uploaded to independently verify the functionality of Digital Pins 7 and 8. After confirming the output ports and LEDs were undamaged, the DHT11 sensor was correctly rewired. A diagnostic code was then run to confirm the sensor survived the short circuit, leading to successful system integration.

Troubleshooting - Hardware Issue 2: Unstable Output Component

- **Problem:** During the alert testing phase, the originally planned active buzzer demonstrated mechanical instability due to loose internal contacts.
- **Solution:** To ensure mechanical stability and reliable demonstration, the output architecture was strategically optimized. The faulty buzzer was replaced by a second LED, successfully transforming the design into a highly stable Dual-LED visual alert system without compromising the multi-level logic.

Future Improvements

1. Integrate an I2C LCD display for real-time temperature visualization.

2. Add current-limiting resistors to protect LEDs during long-term operation.

3. Implement a WiFi module for remote IoT monitoring.

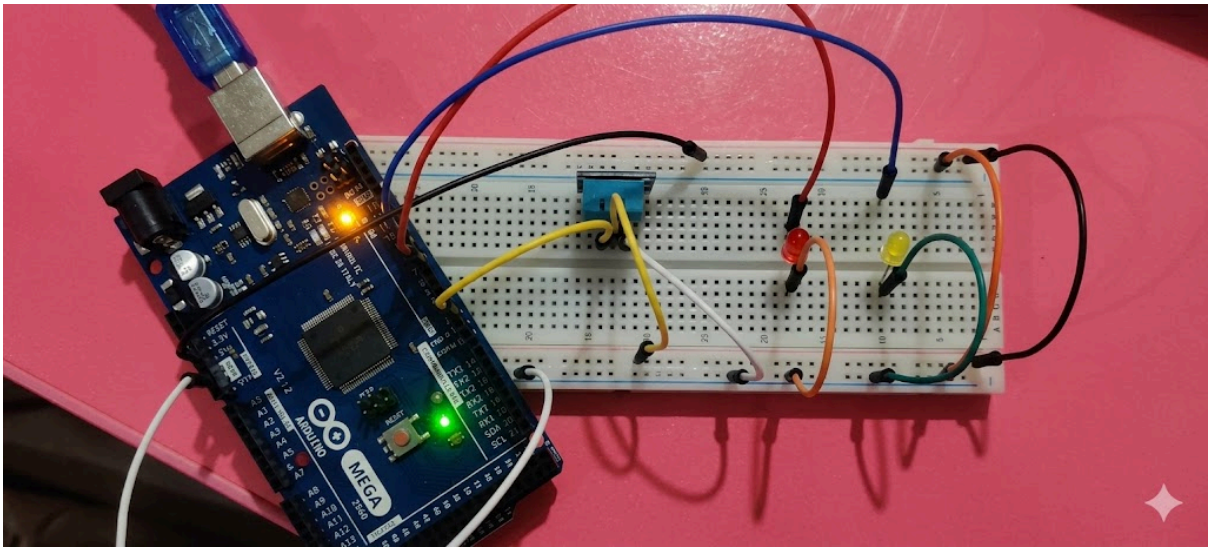
4. Export temperature data for further analysis using MATLAB.

Conclusion

A real-time, multi-level temperature monitoring system was successfully developed and deployed using an Arduino Mega 2560 and a DHT11 sensor. The prototype proved highly responsive to environmental changes, accurately triggering a Warning LED at 32°C and a secondary Alert LED at 34°C. The troubleshooting phase further validated the robustness of the system logic.The developed system demonstrates reliable temperature monitoring and alert functionality. The prototype successfully detected temperature changes and activated appropriate warning indicators. The system can serve as a basic foundation for future IoT-based monitoring systems.

Project Evidence

- **Circuit Photography:**



- **Arduino Source Code:**

```
#include <DHT.h>


#define DHTPIN 2

#define DHTTYPE DHT11
```

```
DHT dht(DHTPIN, DHTTYPE);

// Define pins for the two LEDs
#define LED_WARNING 7  // Warning LED (Turns ON at 32°C)
#define LED_ALERT 8    // Alert LED (Turns ON at 34°C)

// Define temperature thresholds
const float ALERT_TEMP = 34.0;
const float WARNING_TEMP = 32.0;

void setup() {
  Serial.begin(9600);
  Serial.println("Dual-LED Temperature System Starting...");
  dht.begin();

  pinMode(LED_WARNING, OUTPUT);
  pinMode(LED_ALERT, OUTPUT);

  // Ensure both LEDs are OFF on startup
  digitalWrite(LED_WARNING, LOW);
  digitalWrite(LED_ALERT, LOW);
}

void loop() {
  // Wait 2 seconds between measurements (DHT11 requirement)
```

```
delay(2000);

float tempC = dht.readTemperature();

// Error handling if the sensor fails to read
if (isnan(tempC)) {
  Serial.println("Error: Sensor disconnected or reading failed!");
  return;
}

// Output current temperature to Serial Monitor
Serial.print("Temperature: ");
Serial.print(tempC, 1);
Serial.println(" *C");

// --- Dual-LED Trigger Logic ---
if (tempC >= ALERT_TEMP) {
  // High Temperature Alert (>= 34°C): Both LEDs ON
  digitalWrite(LED_WARNING, HIGH);
  digitalWrite(LED_ALERT, HIGH);
  Serial.println("Status: ALERT! (Both LEDs ON)");
}
else if (tempC >= WARNING_TEMP) {
  // Medium Temperature Warning (>= 32°C): Only Warning LED ON
  digitalWrite(LED_WARNING, HIGH);
```

```
    digitalWrite(LED_ALERT, LOW);
    Serial.println("Status: WARNING! (Warning LED ON, Alert LED OFF)");
  }
  else {
    // Normal Temperature (< 32°C): Both LEDs OFF
    digitalWrite(LED_WARNING, LOW);
    digitalWrite(LED_ALERT, LOW);
    Serial.println("Status: Normal (All OFF)");
  }

  Serial.println("--------------------------");
}
```

- **Video Demonstration:**

  https://youtu.be/XToy7RldGQM